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Github Link: <https://github.com/jkwon128/INFO2950-Project>

Overview: We came up with three dataset ideas that all connect inflation to S&P 500 performance. The main difference is how we measure inflation. Looking at a few options now will help us decide which one makes the most sense to use for Phase II.

1. Headline CPI Inflation vs S&P 500 Returns (Primary Idea)

- Does higher headline inflation correspond to lower monthly S&P 500 returns?

Description: For this idea, we would compare U.S. inflation with S&P 500 performance. We would use CPI-U as our measure of inflation and monthly S&P 500 returns as our measure of stock-market performance. This would let us see whether months with higher inflation also tend to have lower stock returns.

Availability and Sources:

- S&P 500 (SP500), daily closing index values, available through FRED. FRED reports that this series contains 10 years of daily history and is sourced from S&P Dow Jones Indices LLC. <https://fred.stlouisfed.org/series/SP500>
- CPI-U, U.S. city average, All items, not seasonally adjusted (series CUUR0000SA0), available from the U.S. Bureau of Labor Statistics. <https://data.bls.gov/timeseries/CUUR0000SA0>
- BLS Public Data API documentation. Version 1 can be used without registration; Version 2 requires a free registration key and allows larger queries. https://www.bls.gov/developers/api_FAQs.htm

Collection and combination plan: We would download the S&P 500 data from FRED and get CPI-U data from the BLS Public Data API. We would probably use data from around 2017 through the most recent month that both sources have. Since the S&P 500 data are daily and CPI is monthly, we would turn the S&P 500 data into monthly returns and calculate the year-over-year inflation rate from CPI. Then we would join the two datasets by year and month, so each row represents one month with both an inflation rate and an S&P 500 return.

Trustworthiness and data quality: We think both sources are very reliable. The S&P 500 data come from S&P Dow Jones Indices and are available through FRED, which is run by the Federal Reserve Bank of St. Louis. CPI-U is produced by the U.S. Bureau of Labor Statistics, which is the main government source for U.S. inflation data. One thing to keep in mind is that the FRED S&P 500 series does not include dividends, so it measures price returns rather than total returns. We also would not claim that inflation causes stock returns to change, since a lot of other factors

can affect the market at the same time.

2. Core CPI Inflation vs S&P 500 Returns

- Is the relationship between inflation and S&P 500 returns different when we leave out food and energy prices?

Description: This idea is similar to Idea 1, but instead of headline CPI, we would use core CPI. Core CPI leaves out food and energy because those prices can move around a lot. We could compare core inflation with S&P 500 returns and see whether this measure has a clearer relationship with stock-market performance than headline inflation.

Availability and Sources:

- S&P 500 (SP500), daily close, from FRED / S&P Dow Jones Indices LLC.
<https://fred.stlouisfed.org/series/SP500>
- CPI-U: All items less food and energy, not seasonally adjusted (series CUUR0000SA0L1E), from the U.S. Bureau of Labor Statistics.
<https://data.bls.gov/timeseries/CUUR0000SA0L1E>
- BLS CPI methods and quality-adjustment documentation.
<https://www.bls.gov/cpi/methods-overview.htm>

Collection and combination plan: We would build the S&P 500 monthly returns the same way as in Idea 1, but use the BLS core CPI series instead of headline CPI. We would calculate year-over-year core inflation and join the two datasets by year and month. This would let us compare stock returns with inflation that excludes food and energy.

Trustworthiness and data quality: FRED and BLS are both well-established sources, and BLS explains how it collects and adjusts CPI data. One possible issue is that core CPI leaves out food and energy even though those costs still matter to consumers and companies. Because of that, we would treat core CPI as another way to measure inflation rather than assuming it is automatically better than headline CPI.

3. Inflation Expectations vs S&P 500 Returns

- Are higher consumer inflation expectations associated with lower S&P 500 returns?

Description: Instead of only looking at inflation that has already happened, this idea would look at what consumers expect inflation to be over the next year. Since investors care about

what might happen in the future, inflation expectations could also be related to stock-market performance. We would compare the University of Michigan one-year inflation expectation measure with monthly S&P 500 returns.

Availability and Sources:

- S&P 500 (SP500), daily close, from FRED / S&P Dow Jones Indices LLC.
<https://fred.stlouisfed.org/series/SP500>
- University of Michigan: Inflation Expectation (MICH), monthly percent, based on the Surveys of Consumers and distributed through FRED.
<https://fred.stlouisfed.org/series/MICH>

Collection and combination plan: We would download both series, convert the S&P 500 data into monthly returns, and join the datasets by year and month. We could first compare inflation expectations with returns from the same month.

Trustworthiness and data quality: The inflation-expectations data come from the University of Michigan Surveys of Consumers, which has been collected for many years, and the series is also available through FRED. The S&P 500 data come from the same source as in the other ideas. One concern is that survey answers can be noisy and may reflect people's overall feelings about the economy, not just what they think inflation will be. The latest observation may also be released on a different schedule from the market data.

Question for reviewers:

1. For our main outcome, is it okay to use monthly returns from the FRED S&P 500 price index, or should we find a version that also includes dividends?
2. Would year-over-year CPI inflation make more sense for this project than month-over-month inflation?
3. When we merge inflation and stock returns, is matching them by calendar month enough, or should we line the data up with the actual CPI release date?
4. Out of headline CPI, core CPI, and inflation expectations, which one would make the most sense as our main inflation measure for Phase II? Could we use the others as extra comparisons?
5. Is about 10 years of monthly data enough for the correlation/regression work in this class, or should we try to find a longer S&P 500 dataset?

Sources:

- Federal Reserve Bank of St. Louis (FRED), S&P 500 series SP500.
<https://fred.stlouisfed.org/series/SP500>
- U.S. Bureau of Labor Statistics, CPI series ID guide and public data tools.
<https://www.bls.gov/cpi/factsheets/cpi-series-ids.htm>
- U.S. Bureau of Labor Statistics, Public Data API FAQ.
https://www.bls.gov/developers/api_FAQs.htm
- U.S. Bureau of Labor Statistics, CPI methods overview.
<https://www.bls.gov/cpi/methods-overview.htm>
- Federal Reserve Bank of St. Louis (FRED), University of Michigan Inflation Expectation series MICH. <https://fred.stlouisfed.org/series/MICH>